# Jarrett D. Phillips, Ph.D.

**School of Computer Science**
**Department of Integrative Biology**
**One Health Institute**
**University of Guelph**
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## RESEARCH AREAS

Computational statistics · Data science · Machine learning · Statistical modelling · Biodiversity informatics · eDNA and DNA barcoding

## ACADEMIC APPOINTMENTS

**Adjunct Professor**, School of Computer Science 2023–Present
University of Guelph

**Affiliated Faculty**, One Health Institute 2026
University of Guelph

## EDUCATION

**Ph.D. in Computational Sciences**, University of Guelph 2022
Co-Advisors: Dr. Daniel Gillis and Dr. Robert Hanner
Advisory Committee Members: Dr. Deborah Stacey and Dr. Graham Taylor
Dissertation: A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Sufficiency: Implications for DNA Barcode Gap Estimation

**Master of Bioinformatics**, University of Guelph 2014
Co-Advisors: Dr. Robert Hanner and Dr. Daniel Ashlock
Major Paper: Assessing DNA Barcode Haplotype Sampling Diversity in the Ray-finned Fishes (Chordata: Actinopterygii)

**BSc. (Hons.)** in Biological Science, University of Guelph 2013
Coursework in bioinformatics, ecology, evolutionary biology, comparative animal physiology, genetics, mathematics, and statistics

## RESEARCH EXPERIENCE

**Postdoctoral Fellow** 2023–Present
University of Guelph
GBADs Informatics Team
Stacey Lab, School of Computer Science
Supervisor: Dr. Deborah Stacey

**Postdoctoral Fellow** 2023–Present
University of Guelph
Gillis Lab, School of Computer Science
Hanner Lab, Department of Integrative Biology
Supervisors: Drs. Daniel Gillis and Robert Hanner

**Postdoctoral Fellow** 2022
University of Guelph
Hanner Lab, Department of Integrative Biology
Supervisor: Dr. Robert Hanner

## RESEARCH FUNDING

### Grants

*Completed*

0. **Canada First Research Excellence Fund (CFREF) Food From Thought Advancing Research Impact Fund (ARIF) – Livestock Innovation Grant** 2024
University of Guelph $40,000 CAD
A Modular Decision- and Policy-Making Tool for Global Livestock Disease Burden Assessment
**Role**: Co-applicant

-1. **Canada First Research Excellence Fund (CFREF) Food From Thought Advancing Research Impact (ARIF) Fund** 2023
University of Guelph $30,000 CAD
Forecasting Seafood Fraud Occurrence in the Supply Chain Within Major Canadian Cities Using Bayesian Hierarchical Binary Logistic Time-Series Regression Modeling
**Role**: HQP

*Submitted*

0. **NSERC Discovery Grant** 2026
University of Guelph Amount $TBD CAD
Computational Statistical Methods for Specimen Sampling Sufficiency and Species Genetic Diversity Assessment
**Role**: Principal Investigator

-1. **NSERC Alliance Society Grant** 2026
University of Guelph $2.5 million CAD
AI-Driven Biodiversity Restoration: Scalable Solutions for Ecosystem Monitoring and Conservation
**Role**: Collaborator

-2. **NSERC CREATE Grant** 2026
University of Guelph $1.65 million CAD
AI-Driven Training of Next-Generation Conservation Scientists
**Role**: Collaborator

*To be Submitted*

0. **Advancing Research Impact Fund: "Quick Wins" Knowledge Mobilization Fund** 2026
University of Guelph $20,000 CAD
GBADs ModelBuilder as a Tool to Teach Livestock Population Dynamics and Disease Burden Risk Estimation
**Role**: Co-applicant

### Awards

0. **SoCS Travel Grant** 2019
University of Guelph $1,000 CAD

-1. **Arthur D. Latornell Graduate Travel Grant** 2019
University of Guelph $500 CAD

-2. **Graduate Teaching Assistantships** 2016–2020
University of Guelph $35,000 CAD

-3. **Graduate Research Assistantships** 2017–2019
University of Guelph $11,000 CAD

### Scholarships

0. **CPES Graduate Dean's Scholarship** 2017
University of Guelph $3,500 CAD

-1. **CPES Graduate Excellence Entrance (GEE) Scholarship** 2016
University of Guelph $30,000 CAD

## REFEREED WORK

**Publication record:** 9 peer-reviewed journal articles (7 first-author, 2 co-author) and 2 book chapters (1 first-author, 1 co-author), plus 3 first-author manuscripts in preprint or to be submitted. Senior (last) author on 2 student-led manuscripts

***** Indicates students under my direct mentorship/supervision
****** Indicates students under my co-mentorship/co-supervision

### Journal Articles

*Published or Accepted*

0. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2026). Swimming in deep uncertainty: How proper statistical modelling can help expose seafood product mislabeling. *CHANCE*, 39(3): 48-56. DOI: 10.1080/09332480.2026.2725526

-1. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2026). Statistical modelling of seafood fraud highlights uncertainties in products from Metro Vancouver, British Columbia, Canada: Revisiting Hu *et al.* (2018). *Journal of Food Science*, 91: e71201. DOI: 10.1111/1750-3841.71201

-2. Raymond, K., Sobkowich, K.E., **Phillips, J.D.**, Nguyen, L., McKechnie, I., Mohideen, R.N., Fitzjohn, W., Szurkowski, M., Davidson, J., Rushton, J., Stacey, D.A., and Bernardo, T.M. (2024). GBADs informatics strategy: User-centric tools, data quality, and model interoperability. *WOAH Scientific and Technical Review*, 43: 96–107. DOI: 10.20506/rst.43.3522.

-3. **Phillips, J.D.**, Athey, T.B.T., McNicholas, P.D., and Hanner, R.H. (2023). VLF: An R package for the analysis of very low frequency variants in DNA sequences. *Biodiversity Data Journal*, 11: e96480. DOI: 10.3897/BDJ.11.e96480.

-4. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2022). Lack of statistical rigor in DNA barcoding likely invalidates the presence of a true species' barcode gap. *Frontiers in Ecology and Evolution*, 10: 859099. DOI: 10.3389/fevo.2022.859099.

-5. D'Ercole, J., Dincă, V., Opler, P.A., Kondla, N.G., Schmidt, C.B., **Phillips, J.D.**, Robbins, R., Burns, J.M., Miller, S.E., Grishin, N., Zakharov, E.V., deWaard, J.R., Ratnasingham, S., and Hebert, P.D.N. (2021). A DNA barcode library for the butterflies of North America. *PeerJ*, 9: e11157. DOI: 10.7717/peerj.11157.

-6. **Phillips, J.D.**, *French, S.H., Hanner, R.H., and Gillis, D.J. (2020). HACSim: An R package to estimate intraspecific sample sizes for genetic diversity assessment using haplotype accumulation curves. *PeerJ Computer Science*, 6(192): 1–37. DOI: 10.7717/peerj-cs.243. *Top 5 most viewed article in category "Optimization Theory and Computation"*.

-7. **Phillips, J.D.**, Gillis, D.J., and Hanner, R.H. (2019). Incomplete estimates of genetic diversity within species: Implications for DNA barcoding. *Ecology and Evolution*, 9(5): 2996–3010. DOI: 10.1002/ece3.4757.

-8. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii). *DNA Barcodes*, 3: 66–73. DOI: 10.1515/dna-2015-0008.

*Preprints*

0. *Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** (2026). Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. bioRxiv. DOI: 10.64898/2026.09.16.752056.

-1. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. (2025). A Bayesian coalescent model of the DNA barcode gap. Authorea. DOI: 10.22541/au.174073683.37707806/v1

-2. **Phillips, J.D.** and *De Vuono-Fraser, F.A. (2024). Statistical modelling of seafood fraud in the Canadian supply chain. bioRxiv. DOI: 10.1101/2024.02.05.578947.

*Submitted or Under Revision*

0. *Toth, N., Antonie, L., Hanner, R.H., Gillis, D.J., and **Phillips, J.D.** Mining association rules for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling. Submitted to *Oikos*.

*To Be Submitted*

0. **Phillips, J.D.**, Hubert, N., and Hanner, R.H. A Bayesian-informed coalescent model of the DNA barcode gap. Targeted to *Methods in Ecology and Evolution*.

*In Preparation (Selected)*

0. *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** The influence of wholesaler fraud and consumer perception on IUU fishing in a model of the seafood supply chain. Targeted to *Proceedings of the National Academy of Sciences*.

-1. Diao, J., Elliott, T.A., **Phillips, J.D.**, You, J., and Adamowicz, S.J. Benchmarking imputation methods for insect trait data across missingness mechanisms and trait types. Targeted to *Methods in Ecology and Evolution*.

-2. D'Ercole, J., Dapporto, L., **Phillips, J.D.**, Dincă, V.E., Vila, R., Talavera, G., and Hebert, P.D.N. Macrogenetics of North American butterflies – The impact of Quaternary climatic fluctuations. Targeted to *Proceedings of the National Academy of Sciences* .

**Book Chapters**

*Published or Accepted*

0. **Phillips, J.D.**, Griswold, C.K., Young, R.G., Hubert, N., and Hanner, R.H. (2024). A Measure of the DNA Barcode Gap for Applied and Basic Research. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY. URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_24

-1. Hubert, N., **Phillips, J.D.**, and Hanner, R.H. (2024). Delimiting Species with Single-Locus DNA Sequences. In: DeSalle, R. (eds) DNA Barcoding. Methods in Molecular Biology, vol 2744. Humana, New York, NY. URL: https://link.springer.com/protocol/10.1007/978-1-0716-3581-0_3

*Submitted or Under Revision*

0. *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain. *VII AMMCS International Conference*, Springer Nature.

## NON-REFEREED WORK

*Ph.D. Thesis*

Phillips, J.D. (2022). A Novel Statistical Framework for Assessment of Intraspecific Haplotype Sampling Sufficiency: Implications for DNA Barcode Gap Estimation. University of Guelph Atrium.
URL: https://atrium.lib.uoguelph.ca/items/8addfcc5-f21c-4691-89b7-c4db051892eb

***** Indicates students under my direct mentorship/supervision

0. *Cui, R.C. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House

-1. *Toth, N. and **Phillips, J.D.** (2026). Student-contributed poster for University of Guelph March Open House

-2. *Cui, R.C. and **Phillips, J.D.** (2025). The Importance in Design as a Computer Scientist. Student-contributed SoCS Show and Tell video URL: https://github.com/RichaCui70/Dynamics-between-Seafood-Fraudsters-and-Buyers/tree/main/misc_resources

-3. **Phillips, J.D.**, *De Vuono-Fraser, F.A., Gillis, D.J., and Hanner, R.H. (2024). Statistical modelling of seafood fraud. Whiteboard explainer video. URL: https://www.youtube.com/watch?v=4jU8cU8M70U

-4. *Toth, N. and **Phillips, J.D.** (2024). eDNA Collection Gets a Tech Update. Student-contributed CEPS Research Highlights article. URL: https://www.uoguelph.ca/ceps/news/2024/07/edna-collection-gets-tech-upgrade

-5. *Toth, N. and **Phillips, J.D.** (2024). Unravelling eDNA with Association Rule Mining. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: https://danielgillis.wordpress.com/2024/08/15/unravelling-edna-with-association-rule-mining/

-6. **Phillips, J.D.** (2024). Summer URA Position. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: https://danielgillis.wordpress.com/2024/02/07/summer-ura-position-2/

-7. **Phillips, J.D.** (2022). Mind the Gap – The DNA Barcode Gap, That Is. Contributed CEPS Research Highlights article. URL: https://www.uoguelph.ca/ceps/news/2022/08/mind-gap-%E2%80%93-dna-barcode-gap

-8. **Phillips, J.D.** (2020). Barcode Cracking. Contributed CEPS Research Highlights article. URL: https://www.uoguelph.ca/ceps/news/2020/02/barcode-cracking

-9. **Phillips, J.D.** (2020). Protecting Biodiversity Through the Lens of Genetic Diversity. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: https://danielgillis.wordpress.com/2020/01/30/protecting-biodiversity-through-the-lens-of-genetic-diversity/

-10. **Phillips, J.D.** (2019). IBOL8 and the Midnight Sun. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: https://danielgillis.wordpress.com/2019/07/02/reflections-ibol8-and-the-midnight-sun/

-11. **Phillips, J.D.** (2017). The Big Five and IBOL7. Contributed guest post to the Science Borealis-syndicated blog of Dr. Daniel Gillis. URL: https://danielgillis.wordpress.com/2017/12/06/reflections-the-big-five-and-ibol7/

-12. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding: Are current sampling levels enough? Contributed guest post to the DNA Barcoding Blog of Dr. Dirk Steinke. URL: http://dna-barcoding.blogspot.com/2016/01/guest-post-sample-size-estimation-for.html

-13. **Phillips, J.D.** (2016). Sample size estimation for DNA barcoding of ray-finned fishes: Are current sampling levels enough? Contributed newsletter article to the Barcode Bulletin, 7(1). URL: https://journal.lib.uoguelph.ca/index.php/ibol/issue/view/345

## CONFERENCE PRESENTATION AND PROCEEDINGS

***** Indicates students under my direct mentorship/supervision

0. **The VII AMMCS International Conference** (student oral presentation) 2026
   Wilfrid Laurier University, Canada
   *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** · Modelling the Interplay between IUU Fishing, Wholesaler Fraud, and Consumer Awareness in the Seafood Supply Chain

-1. **Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) International Conference** (oral presentation) 2026
   University of Guelph, Canada
   **Phillips, J.D.** · Association rule mining for targeted spatiotemporal aquatic environmental DNA (eDNA) sampling

-2. **Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) International Conference** (student poster presentation) 2026
   University of Guelph, Canada
   *Toth, N., Antonie, M.L., and **Phillips, J.D.**. · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets

-3. **39th Canadian Conference on Artificial Intelligence** (short paper and student poster presentation) 2026
   Simon Fraser University, Canada
   *Toth, N., Antonie, M.L. and **Phillips, J.D.** · To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets.

-4. **SoCS Undergraduate Summer Project Show and Tell** (student oral presentation) 2025
   University of Guelph, Canada
   *Cui, R.C., Yodzis, M.P., and **Phillips, J.D.** · The Importance in Design as a Computer Scientist

-5. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2024
   University of Guelph, Canada
   *Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets

-6. **CBS Undergraduate Poster Session** (student poster presentation) 2024
   University of Guelph, Canada
   *Toth N. and **Phillips, J.D.** · Association Rule Mining of eDNA Datasets

-7. **9th International Barcode of Life Conference** (poster presentation) 2024
   Estação das Docas, Belém, Brazil
   **Phillips, J.D.** · A Measure of the DNA Barcode Gap for Applied and Basic Research (*abstract accepted, not attended*)

-8. **GBADs Informatics Technical Workshop** (oral presentation) 2023
   University of Liverpool, England
   **Phillips, J.D.** · The GBADs R Package (and Why We Need It!)

-9. **CEPS Student Research Day** (student poster presentation) 2023
   University of Guelph, Canada
   *De Vuono-Fraser, F.A. and **Phillips, J.D.** · Estimating Seafood Mislabelling Rates in Canada Using Bayesian Modelling

-10. **Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) International Conference** (poster presentation) 2023
   University of Guelph, Canada
   **Phillips, J.D.** · Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes

-11. **8th International Barcode of Life Conference** (oral presentation) 2019
NTNU University Museum and Norwegian Biodiversity Information Centre, Norway
**Phillips, J.D.** · HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness

-12. **Guelph BioMathematics and Statistics (BioM&S) Symposium** 2019
**Artificial Intelligence and Machine Learning in Biology** (attended)
University of Guelph, Canada

-13. **CEPS Undergraduate Student Poster Day** (student poster presentation) 2018
University of Guelph, Canada
*French, S.H. and **Phillips, J.D.** · Estimating Sampling Size Using Haplotype Accumulation Curves and Semiparametric Models

-14. **7th International Barcode of Life Conference** (oral presentation) 2017
University of Johannesburg, South Africa
**Phillips, J.D.** · Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough?

-15. **6th International Barcode of Life Conference** (poster presentation) 2015
University of Guelph, Canada
**Phillips, J.D.** · An exploration of sufficient sampling effort to describe intraspecific DNA barcode haplotype diversity: examples from the ray-finned fishes (Chordata: Actinopterygii).

### Conference Proceedings

*Published or Accepted*

0. *Toth, N., Antonie, M.L., and **Phillips, J.D.** To correct or not to correct?: Assessing the multiple comparisons problem for association rule mining of environmental DNA (eDNA) detection survey datasets. *39th Canadian Conference on Artificial Intelligence*, **318**: 1012–1019. Short paper and poster.

-1. Morey, K., Loeza-Quintana, T., **Phillips, J.**, and Hanner R. (2023). Haplotype diversity reveals challenges and opportunities for developing targeted detection assays for *COI* in Canadian freshwater fishes. Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference. Poster.

-2. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2019). HACSim: Iterative extrapolation of haplotype accumulation curves for assessment of intraspecific COI DNA barcode sampling completeness. Scientific abstracts from the 8th International Barcode of Life Conference, Trondheim, Norway (ed. Torbjørn Ekrem), *Genome*, 62(6): 349–453. Oral presentation.

-3. **Phillips, J.D.**, Gillis, D., and Hanner, R. (2017). Intraspecific sample size estimation for DNA barcoding: Are current sampling levels enough? Scientific abstracts from the 7th International Barcode of Life Conference, Johannesburg, South Africa (ed. M. van der Bank), *Genome*, 60(11): 881–1019. Oral presentation.

-4. **Phillips, J.D.**, Gwiazdowski, R.A., Ashlock, D., and Hanner, R. (2015). An exploration of sufficient sampling effort to describe intraspecific haplotype diversity in the ray-finned fishes (Chordata: Actinopterygii). Scientific abstracts from the 6th International Barcode of Life Conference, Guelph, ON., Canada (ed. S.J. Adamowicz), *Genome*, 58(5): 163–303. Poster.

## SOFTWARE AND COMPUTATIONAL TOOLS

98,000+ software downloads through the Comprehensive R Archive Network (CRAN)

0. **Seafood Fraud Modelling App** · Python web application · URL: https://dynamics-between-seafood-fraudsters-and-buyers.streamlit.app

-1. **GBADs ModelBuilder** · R Shiny web application · URL: https://animalhealthmetrics.org/2025/10/14/introduction-to-gbads-new-model-builder-prototype/

-2. **RulesTools** · R package · >3000 downloads · DOI: 10.32614/CRAN.package.RulesTools

-3. **Frequentist/Bayesian DNA barcode gap coalescent-informed estimators** · R scripts
-4. **VLF** (**V**ery **L**ow **F**requency) · R package · >40000 downloads · DOI: 10.32614/CRAN.package.VLF
-5. **HACSim** (**H**aplotype **A**ccumulation **C**urve **Sim**ulator) · R package · R Shiny web application · >50000 downloads · DOI: 10.32614/CRAN.package.HACSim

## STUDENT SUPERVISION AND MENTORSHIP

Mentored/co-supervised **18 undergraduate and graduate students** across computer science, bioinformatics, statistics, and biology.

* Indicates students under my direct mentorship/supervision
** Indicates students under my co-mentorship/co-supervision

*Undergraduate*

***Brendan Carl Rosario** 2025
Undergraduate Student Volunteer · Supervised Machine Learning for eDNA spatiotemporal sampling

***Richard Cui** 2025
Summer Undergraduate Research Assistant (URA) · Dynamical modelling of seafood fraud in the supply chain

***Nikolett Toth** 2024–2025
CIS*4900/4910 · Mining association rules for eDNA spatiotemporal sampling

***Nikolett Toth** 2024
Summer Undergraduate Research Assistant (URA) · Mining association rules for eDNA spatiotemporal sampling

***Fynn De Vuono-Fraser** 2024
CIS*4900/4910 · Bayesian modelling of seafood fraud in the Canadian supply chain

***Zaid Al-Gayyali** 2023
Summer Undergraduate Research Assistant (URA) · Seafood Fraud Visualization Tool Shiny app

***Fynn De Vuono-Fraser** 2023
STAT*4600 · Bayesian modelling of seafood fraud in the Canadian supply chain

***Navdeep Singh** 2021
CIS*4900 · HACSim R Shiny web application

***Scarlett Bootsma** 2020–2021
CIS*4900/4910 · HACSim simulation study

****Maya Persram** 2020
Hanner Lab volunteer · R reporting ecological meta-analysis

****Ashley Chen** 2020
Hanner Lab volunteer · R reporting ecological meta-analysis

****Olivia Friesen Kroeker** 2020
Hanner Lab volunteer · R reporting ecological meta-analysis

***Steven French** 2018
CIS*4900/4910 · HACSim R package

****Julia Harvie** 2018–2019
MCB*4500/4510 · Data mining GenBank and BOLD

*Graduate*

****Nathan Zeinstra** 2024
IBIO*6070 · Bayesian habitat occupancy modelling for sea lamprey detection using eDNA

****Amina Asif** 2022
BINF*6999 · DNA barcode gap analysis of Canadian disease vectors and agricultural pests

****Danielle St. Jean** (*withdrawn*) 2018–2019
MSc. thesis (Mathematics) · DNA barcode sequence classification with machine learning

****Christina Fragel** 2018–2019
BINF*6999 · DNA barcode sequence classification with machine learning

****Jiaojia (Paula) Yu** 2018–2019
BINF*6999 · MDMAPR R Shiny app

****Ankita Bhanderi** 2018
BINF*6999 · Data mining GenBank and BOLD

## TEACHING EXPERIENCE

**Guest Lecturer** 2026
University of Guelph
**CIS*4020 – Data Science**
Talk title: Association Rule Mining: From Market Baskets to Environmrntal DNA (eDNA)

**Course Instructor** 2025
University of Guelph
**CIS*1910 – Discrete Structures in Computing I**

**Graduate Teaching Assistant (GTA)** 2016–2020
University of Guelph
**CIS*3130 – System Modelling and Simulation** 2020
**CIS*1910 – Discrete Structures in Computing I** 2017
**CIS*2460 – Modelling of Computer Systems** 2016–2019

## ACADEMIC SERVICE

0. **Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference** 2026
University of Guelph
   - Conference organizer and volunteer

-1. **39th Canadian Conference on Artificial Intelligence** 2026
Simon Fraser University
   - Program committee reviewer for Long and Short Papers Track

-2. **GBADs Informatics Technical Workshop** 2023
University of Liverpool
   - Workshop organizer

-3. **Pathway to Increase Standards and Competency in eDNA Surveys (PISCeS) Conference** 2023
University of Guelph
   - Conference organizer and volunteer

-4. **School of Computer Science (SoCS) Faculty Search Committee** 2018
University of Guelph
   - Member of hiring panel for Associate Professor in Cybersecurity

-5. **School of Computer Science (SoCS) Faculty Search Committee** 2017–2018
University of Guelph
   - Member of hiring panel for 2-year Contractually Limited Assistant Professor in Cybersecurity

## ACADEMIC PEER REVIEW SERVICE

*Ecology and Evolution*, *F1000Research*, *Frontiers in Ecology and Evolution*, *Lifestyle Genomics*, *Methods in Ecology and Evolution*, *Mitochondrial DNA Part A*, *Molecular Biology Reports*, *Molecular Ecology Resources*, *Nature Communications*

## VOLUNTEER EXPERIENCE

**CCMPS Research Day & Student Poster Presentations** 2026
University of Guelph

- Served as judge for undergraduate and graduate student posters

**CEPS Student Research Connections Networking Night** 2025
University of Guelph

- Connected with undergraduate students for summer URA project recruitment

**CIS*3750 Wireframing session** 2021–Present
University of Guelph

- Graded students on mobile app prototypes for various community partners using Qualtrics surveys